

Empirical Methods for Policy Evaluation

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Randomized Experiments and Dynamic Latent Factor Models

① Randomized control trials (RCTs)

- ⇒ SUTVA and assignment mechanisms
- ⇒ Randomization inference
- ⇒ Randomization designs
- ⇒ Stratified RCTs and Clustered RCTs

② Dynamic Latent Factor Models

- ⇒ Cunha-Nielsen-Williams (2021, ARE)

③ The production of human capital in the early years

- ⇒ [Attanasio-Cattan-Fitzsimons-Meghir-Rubio \(AER, 2020\)](#)

SUTVA and assignment mechanisms

Causal Inference as a Missing Data Problem

- Population of units, indexed by $i = 1, \dots, N$
- Treatment indicator W_i taking values 0 and 1
- For $i \in \{1, \dots, N\} \exists$ one realized outcome and one missing potential outcome

$$Y_i^{\text{obs}} = Y_i(W_i) = \begin{cases} Y_i(0) & \text{if } W_i = 0 \\ Y_i(1) & \text{if } W_i = 1 \end{cases}$$

$$Y_i^{\text{miss}} = Y_i(1 - W_i) = \begin{cases} Y_i(1) & \text{if } W_i = 0 \\ Y_i(0) & \text{if } W_i = 1 \end{cases}$$

- Unit-level causal effect $Y_i(1) - Y_i(0)$ is unobserved

The Stable Unit Treatment Value Assumption (SUTVA)

- We will generally need to predict, or impute, the missing potential outcome
- Denote $W_{-i} = (W_j)_{j \neq i}$, SUTVA requires that

$$(Y_i(1), Y_i(0)) \perp W_{-i}$$

1 No interference

$\Rightarrow$ Potential outcomes for unit i do not vary with the treatment status of any other j

2 Scale Invariance

$\Rightarrow$ There are no different forms of each treatment, which lead to $\neq$ potential outcomes

Sources of Possible Violations of SUTVA

1 Spillovers and equilibrium effects. E.g.:

- ⇒ Fertilizer in one plot may affect yields in contiguous plots
- ⇒ Immunization efficacy may depend on the number of people immunized
- ⇒ Prob(job) after training may be affected by the number of people trained

2 Hidden variations of treatments. E.g.:

- ⇒ Endogenous compliance to treatment assignment
- ⇒ Unobserved differences in the method of administering the treatment

Assignment Mechanism

- N units, set $\mathbb{W} = \{0, 1\}^N$ of N -vectors with all elements equal to 0 or 1
- The **assignment mechanism** is a function $P(\mathbf{W}|\mathbf{Y}(0), \mathbf{Y}(1)) \in [0, 1]$ such that

$$\sum_{\mathbf{W} \in \{0, 1\}^N} P(\mathbf{W}|\mathbf{Y}(0), \mathbf{Y}(1)) = 1$$

⇒ The unit-level **assignment probability** is

$$p_i(\mathbf{Y}(0), \mathbf{Y}(1)) = \sum_{\mathbf{W}: W_i=1} P(\mathbf{W}|\mathbf{Y}(0), \mathbf{Y}(1))$$

⇒ The **propensity score** is

$$e(x) = \frac{1}{N(x)} \sum_{i: X_i=x} p_i(\mathbf{X}, \mathbf{Y}(0), \mathbf{Y}(1))$$

Example of Assignment Mechanism 1

- Two units, and so (2^2) possible values for $\mathbf{W}$

$$\mathbf{W} \in \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$$

⇒ The assignment mechanism is equal to $P(\mathbf{W}|\mathbf{Y}(0), \mathbf{Y}(1)) = \frac{1}{4}$

⇒ Unit assignment probability $p_i = p = \frac{1}{2}$ for $i = 1, 2$

⇒ No covariates, and so propensity score $e = p = \frac{1}{2}$

Example of Assignment Mechanism 2

- Two units where only assignments with one treated and one control unit

⇒ The assignment mechanism is

$$P(\mathbf{W}|\mathbf{Y}(0), \mathbf{Y}(1)) = \begin{cases} 1/2 & \text{if } \mathbf{W} \in \left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\} \\ 0 & \text{if } \mathbf{W} \in \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\} \end{cases}$$

⇒ As before, $e = p = \frac{1}{2}$

Regular Assignment Mechanisms

1 Individualistic

$$p_i(\mathbf{Y}(0), \mathbf{Y}(1)) = q(Y_i(0), Y_i(1)), q(\cdot) \in [0, 1]$$

2 Probabilistic

$$0 < p_i(\mathbf{Y}(0), \mathbf{Y}(1)) < 1$$

3 Unconfounded

$$P(\mathbf{W}|\mathbf{Y}(0), \mathbf{Y}(1)) = P(\mathbf{W}|\mathbf{Y}'(0), \mathbf{Y}'(1)) = P(\mathbf{W})$$

Randomized Experiments and Observational Studies

- Individualistic + unconfounded implies

$$P(\mathbf{Y}(0), \mathbf{Y}(1)) = c \cdot \prod_{i=1}^N q^{W_i} (1 - q)^{1 - W_i}$$

⇒ The constant c ensures that the probabilities add to unity

⇒ In this case, $p_i = e = q$

- A regular assignment mechanism with **known and controlled** $q(\cdot)$ is a randomized experiment
- When the **functional form of the assignment mechanism is unknown** we have an observational study

Randomization Inference

Two Approaches for Inference (Not Just in RCTs)

① Asymptotic inference (model-based)

- ⇒ (semi-) **Parametric** models for the conditional mean of observed outcomes
- ⇒ **Observed outcomes** vary through random sampling from a population of units
- ⇒ Compute the distribution of the test statistic through **large-sample properties**

② Randomization inference (design-based)

- ⇒ **Nonparametric**; i.e. no restrictions on the distribution of the potential outcomes
- ⇒ **Treatment assignment** is the sole source of randomness; observed outcomes are fixed
- ⇒ The **assignment mechanism** determines the distribution of the test statistic

Illustrative Example

Unit	Potential Outcomes				
	Cough Frequency (cfa)		Observed Variables		
	$Y_i(0)$	$Y_i(1)$	W_i	X_i (cfp)	Y_i^{obs} (cfa)
1	?	3	1	4	3
2	?	5	1	6	5
3	?	0	1	4	0
4	4	?	0	4	4
5	0	?	0	1	0
6	1	?	0	5	1

The Sharp Null

- The p -value for the **sharp** null hypothesis that the treatment had no effect on cough frequency is

$$H_0 : Y_i(0) = Y_i(1) \forall i = 1, \dots, 6.$$

$\Rightarrow$ The average null $\mathbb{E}[Y_i(0) - Y_i(1)] = 0$ is **weaker than the sharp null** hypothesis

- Under H_0 , all missings in **potential outcomes inferred from observed outcomes**
- One **test statistics** from example above is

$$\begin{aligned} T(W, Y^{\text{obs}}) &= | \bar{Y}_t^{\text{obs}} - \bar{Y}_c^{\text{obs}} | \\ &= | (Y_1^{\text{obs}} + Y_2^{\text{obs}} + Y_3^{\text{obs}})/3 - (Y_4^{\text{obs}} + Y_5^{\text{obs}} + Y_6^{\text{obs}})/3 | \\ &= | 8/3 - 5/3 | = 1.00 \end{aligned}$$

Treatment Assignments Generate Test Distribution

- We can re-do this for $\binom{6}{3} = 20$ **permutations of treatment assignments**
⇒ E.g. instead of $W^{\text{obs}} = (1, 1, 1, 0, 0, 0)$ take $\tilde{W} = (0, 1, 1, 0, 0, 1)$
- **No change in observed outcomes** since $Y_i(0) = Y_i(1) = Y_i^{\text{obs}}$ under H_0
- The value of the test statistic for $\tilde{W}$ is

$$\begin{aligned} T(\tilde{W}, Y^{\text{obs}}) &= | (Y_2^{\text{obs}} + Y_3^{\text{obs}} + Y_6^{\text{obs}})/3 - (Y_1^{\text{obs}} + Y_4^{\text{obs}} + Y_5^{\text{obs}})/3 | \\ &= | 6/3 - 7/3 | = 0.33 \end{aligned}$$

Exact p -values

- Calculate the value of the statistic for each assignment vector
 - ⇒ In previous example, each assignment vector has prior probability=1/20
- How unusual is $T(W, Y^{\text{obs}}) = 1.00$ under the sharp null hypothesis?
- There are 16/20 assignments with $T(\tilde{W}, Y^{\text{obs}}) > T(W, Y^{\text{obs}})$: $p\text{-value} = 0.80$
- In this example, the observed difference could well be due to chance

Approximate p -values

- Recall that the number of distinct values of the treatment vector is $\binom{N_c + N_t}{N_t}$
 - ⇒ For instance, if $N = 100$ and $q = 0.5$ then $\dim(\mathbb{W}^+) = e^{29}$
- We thus need to rely on **numerical approximations** to calculate the p -value
 - ⇒ Draw an N -dimensional vector with N_c zeros and N_t ones from $\mathbb{W}^+$
 - ⇒ Repeat this process $K - 1$ times and approximate the p -value by:

$$\hat{p} = \frac{1}{K} \sum_{k=1}^K \mathbf{1}_{T^{\text{dif},k} \geq T^{\text{dif},\text{obs}}}$$

- ⇒ With $K > 1,000$ each assignment has a similar probability with and without replacement

Randomization designs

Classic Randomization Designs

- Classic randomization designs can be characterized by **restrictions on $\mathbb{W}^+$**

Type of Experiment and Design	Number of Possible Assignments Cardinality of $\mathbb{W}^+$	Number of Units (N) in Sample			
		4	8	16	32
Bernoulli trial	2^N	16	256	65,536	4.2×10^9
Completely randomized experiment	$\binom{N}{N/2}$	6	70	12,870	0.6×10^9
Stratified randomized experiment	$\left(\binom{N/2}{N/4}\right)^2$	4	36	4,900	0.2×10^9
Paired randomized experiment	$2^{N/2}$	4	16	256	65,536

Bernoulli Trials (Coin Tossing)

- $p = e = 0.5$, $\mathbb{W}^+ = \{0, 1\}^N$, $\dim(\mathbb{W}^+) = 2^N$

$$P(\mathbf{W}|\mathbf{X}, \mathbf{Y}(0), \mathbf{Y}(1)) = 0.5^N$$

- More generally, with probability of assignment to treatment $\neq 0.5$

$$P(\mathbf{W}|\mathbf{X}, \mathbf{Y}(0), \mathbf{Y}(1)) = q^{N_t}(1 - q)^{N_c}$$

- No way to ensure “enough” treated and control units under each assignment

Completely Randomized Experiments

- Draw N_t units at random, such that $1 \leq N_t \leq N - 1$
- $q = \frac{N_t}{N}$, and $\dim(\mathbb{W}^+) = \binom{N}{N_t}$
- The set of possible assignment vectors is

$$\mathbb{W}^+ = \left\{ \mathbf{w} \in \mathbb{W} \mid \sum_{i=1}^N W_i = N_t \right\}$$

- Possible **issue with covariate unbalancedness** after treatment assignment

Stratified Randomized Experiments

- The population of units is first partitioned into **blocks or strata** $B_i = B(\mathbf{X}_i)$
- Within each block, we conduct a completely randomized experiment
- The set of possible assignment vectors is

$$\mathbb{W}^+ = \left\{ \mathbf{w} \in \mathbb{W} \mid \sum_{i: B_i = j}^N W_i = N_t(j) \right\}$$

- **More precise inference** as units of same type are in different treat arms

Paired Randomized Experiments

- As many units as treatments within each block
- $N(j) = 2$ and $N_t(j) = 1$ for $j = 1, \dots, N/2$, so that

$$\mathbb{W}^+ = \left\{ \mathbf{w} \in \mathbb{W} \mid \sum_{i: B_i = j}^N W_i = 1 \right\}.$$

$\Rightarrow$ Useful design when N is small and/or J is large

Other Randomization Designs

- These experimental designs have become popular in recent years
 - Clustered randomized experiments
 - ⇒ Partial population designs
 - ⇒ Randomized saturation designs
 - Adaptive randomized experiments

Clustered Randomized Experiments

- Clusters are defined by partitioning the covariate space $G_{ig} = G(\mathbf{X}_i)$
- $\overline{W}_g = \sum_{i:G_{ig}=1} \frac{W_i}{N_g} \in \{0, 1\}$ is the average value of W_i for units in cluster g
- The assignment mechanisms concerns groups of units (clusters)

$$\mathbb{W}^+ = \left\{ \mathbf{W} \in \mathbb{W} \mid \sum_{g=1}^G \overline{W}_g = G_t \right\}$$

- ⇒ Useful design when there are **local interactions between units**
- ⇒ Relax SUTVA within clusters, but maintain it across clusters

Partial Population Designs

- Within-cluster (non-random) program assignment $E_{ig} = \{0, 1\}$
- Write potential outcomes as $Y_i(W_g, E_{ig})$

$$\begin{aligned}ATE &= \mathbb{E}(Y_i(1, 1) - Y_i(0, 1)) = \\&= \mathbb{E}(Y_i | W_g = 1, E_{ig} = 1) - \mathbb{E}(Y_i | W_g = 0, E_{ig} = 1) \\ITE &= \mathbb{E}(Y_i(1, 0) - Y_i(0, 0)) = \\&= \mathbb{E}(Y_i | W_g = 1, E_{ig} = 0) - \mathbb{E}(Y_i | W_g = 0, E_{ig} = 0)\end{aligned}$$

$\Rightarrow$ ATE and ITE depend on $S_g = \sum_{i \in g} E_{i,g}$, which is endogenous

Randomized Saturation Design

- A variant of clustered randomization where
 - ⇒ Assign each cluster to a treatment saturation, $S_g = \sum_{i \in g} W_{ig} \in [0, 1]$
 - ⇒ Assign each individual to a treatment status $W_{ig} = \{0, 1\}$ according to S_g
- Write potential outcomes as $Y_i(W_{ig}, S_g)$

$$\begin{aligned}ATE(s) &= \mathbb{E}(Y_i(1, s) - Y_i(0, 0)) = \\&\quad + \mathbb{E}(Y_i|W_{ig} = 1, S_g = s) - \mathbb{E}(Y_i|W_{ig} = 1, S_g = 0) \\ITE(s) &= \mathbb{E}(Y_i(0, s) - Y_i(0, 0)) = \\&= \mathbb{E}(Y_i|W_{ig} = 0, S_g = s) - \mathbb{E}(Y_i|W_{ig} = 0, S_g = 0)\end{aligned}$$

⇒ Optimal combination of clustered and stratified designs

Adaptive Randomized Experiments

- Waves $t = 1, \dots, T$, sample sizes N_t
- Treatment $D \in \{1, \dots, k\}$, outcomes $Y \in [0, 1]$, covariates X
- Potential outcomes Y^d , and $\theta^{dx} = E[Y_{it}^d | X_{it} = x]$
- Repeated cross-sections: $(Y_{it}^1, \dots, Y_{it}^k; X_{it})$ are i.i.d. across both i and t

⇒ Form **posterior beliefs** P_t over θ , e.g. (thompson sampling):

$$p_t^{dx} = P_t \left(d = \operatorname{argmax}_{d'} \theta^{d'x} \right)$$

⇒ **Stop** when $p_t^{dx} > q$ for $D = d$

Stratified randomized experiments

The Structure of Stratified Randomized Experiments

- Let J be the number of **strata/blocks**, and $N(j), N_c(j), N_t(j)$
- Let $S_i \in \{1, \dots, J\}$ be the stratum for unit i
- Let $B_i(j) = \mathbf{1}_{S_i=j}$ be the stratum indicator for unit i
- The assignment mechanism is

$$P(\mathbf{W} | \mathbf{B}, \mathbf{Y}(0), \mathbf{Y}(1)) = \prod_{j=1}^J \binom{N(j)}{N_t(j)}^{-1} \text{ for } \mathbf{W} \in \mathbb{W}^+$$

$$\Rightarrow \mathbb{W}^+ = \{\mathbf{W} \in \mathbb{W} \mid \sum_{i=1}^N B_i(j) \cdot W_i = N_t(j) \text{ for } j = 1, \dots, J\}$$

Example: Tennessee Project Star

School/ Stratum	No. of Classes	Regular Classes ($W_i = 0$)	Small Classes ($W_i = 1$)
1	4	-0.197, 0.236	0.165, 0.321
2	4	0.117, 1.190	0.918, -0.202
3	5	-0.496, 0.225	0.341, 0.561, -0.059
4	4	-1.104, -0.956	-0.024, -0.450
5	4	-0.126, 0.106	-0.258, -0.083
6	4	-0.597, -0.495	1.151, 0.707
7	4	0.685, 0.270	0.077, 0.371
8	6	-0.934, -0.633	-0.870, -0.496, -0.444, 0.392
9	4	-0.891, -0.856	-0.568, -1.189
10	4	-0.473, -0.807	-0.727, -0.580
11	4	-0.383, 0.313	-0.533, 0.458
12	5	0.474, 0.140	1.001, 0.102, 0.484
13	4	0.205, 0.296	0.855, 0.509
14	4	0.742, 0.175	0.618, 0.978
15	4	-0.434, -0.293	-0.545, 0.234
16	4	0.355, -0.130	-0.240, -0.150
Average		-0.13	0.09
(S.D.)		(0.56)	(0.61)

The Benefits of Stratification

- Consider a DGP with $i = 1, \dots, N$ and one covariate $X_i \in \{0, 1\}$

$$\Rightarrow p(X_i = 1) = p = \frac{1}{2}$$

- Completely randomized design

$$\Rightarrow N_t = qN \text{ and } N_c = (1 - q)N$$

- ATE and its sampling variance

$$\hat{\tau}^{\text{dif}} = \bar{Y}_t^{\text{obs}} - \bar{Y}_c^{\text{obs}}$$
$$\mathbb{V}(\hat{\tau}^{\text{dif}}) = \frac{\sigma_t^2}{N_t} + \frac{\sigma_c^2}{N_c}$$

The Benefits of Stratification

- Stratified design

$$\Rightarrow N_t(f) = pqN \text{ and } N_c(f) = p(1 - q)N$$

$$\Rightarrow N_t(m) = (1 - p)qN \text{ and } N_c(m) = (1 - p)(1 - q)N$$

- CATE (G_i)

$$\hat{\tau}^{\text{dif}}(f) = \bar{Y}_t^{\text{obs}}(f) - \bar{Y}_c^{\text{obs}}(f)$$

$$\hat{\tau}^{\text{dif}}(m) = \bar{Y}_t^{\text{obs}}(m) - \bar{Y}_c^{\text{obs}}(m)$$

- ATE and its sampling variance

$$\hat{\tau}^{\text{strat}} = p\hat{\tau}^{\text{dif}}(f) + (1 - p)\hat{\tau}^{\text{dif}}(m)$$

$$\mathbb{V}(\hat{\tau}^{\text{strat}}) = \frac{p}{N} \left(\frac{\sigma_t^2(f)}{p} + \frac{\sigma_c^2(f)}{1 - p} \right) + \frac{1 - p}{N} \left(\frac{\sigma_t^2(m)}{p} + \frac{\sigma_c^2(m)}{1 - p} \right)$$

The Benefits of Stratification

- The difference in the two variances is

$$\mathbb{V}(\hat{\tau}^{\text{dif}}) - \mathbb{V}(\hat{\tau}^{\text{strat}}) = \frac{p(1-p)}{N} ((\mu_c(f) - \mu_c(m))^2 + (\mu_t(f) - \mu_t(m))^2) \geq 0$$

⇒ The extra-variance in dif vs. strat comes from the **unbalancedness of G_i**

⇒ In principle, $\hat{V}^{\text{dif}} < \hat{V}^{\text{strat}}$ (within-stratum potential outcome variances)

Randomization Inference for Stratified Experiments

- Sharp null hypothesis

$$H_0 : Y_i(1) = Y_i(0) \forall i = 1, 2, \dots, N.$$

- Define average observed outcomes in stratum j as

$$\bar{Y}_t^{\text{obs}}(j) = \frac{1}{N_t(j)} \sum_{i:S_i=j} W_i Y_i^{\text{obs}}$$

$$\bar{Y}_c^{\text{obs}}(j) = \frac{1}{N_c(j)} \sum_{i:S_i=j} (1 - W_i) Y_i^{\text{obs}}$$

⇒ Strata-level propensity score is

$$e(j) = \frac{N_t(j)}{N(j)}$$

Test Statistics

- Within-stratum test statistic

$$T^{\text{dif}}(j) = |\bar{Y}_t^{\text{obs}}(j) - \bar{Y}_c^{\text{obs}}(j)|$$

⇒ Not very informative as we are interested in treatment effects across all strata

- Linear combination of the within-stratum statistics

$$T^{\text{dif}, \lambda_{RSS}} = \left| \sum_{j=1}^J \frac{N_j}{N} \left(\bar{Y}_t^{\text{obs}}(j) - \bar{Y}_c^{\text{obs}}(j) \right) \right|$$

⇒ Need $e(j)$ to vary across j for the test to have power over T^{dif}

Randomization Inference of the Tennessee Project Star

- $B_i(j), i = 1, \dots, 68$ (class-level data)

$$H_0 : Y_i(1) = Y_i(0) \forall i = 1, 2, \dots, 68.$$

- Total number of possible assignments is $(6^{13}) \times 10^2 \times 15$
 - $\Rightarrow$ 13 Schools with two classes in each group: $\binom{4}{2} = 6$
 - $\Rightarrow$ 2 Schools with three small classes and two regular classes: $\binom{5}{2} = 10$
 - $\Rightarrow$ 1 School with four small classes and two regular classes: $\binom{6}{2} = 15$

$$\Rightarrow T^{\text{dif}} = |\bar{Y}_t^{\text{obs}} - \bar{Y}_c^{\text{obs}}| = 0.224, \text{ with } p = 0.034$$

$$\Rightarrow T^{\text{dif}, \lambda_{RSS}} = \left| \sum_{j=1}^J \frac{N_j}{N} (\bar{Y}_t^{\text{obs}}(j) - \bar{Y}_c^{\text{obs}}(j)) \right| = 0.241, \text{ with } p = 0.023$$

Regression Analysis

$$Y_i^{\text{obs}} = \tau W_i + \sum_{j=1}^J \beta(j) B_i(j) + \epsilon_i$$

- Recall that $B_i(j) = \mathbf{1}_{S_i=j}$ is the stratum indicator for unit i

$\Rightarrow \hat{\tau}^{\text{ols}}$ is not a consistent estimator of ATE if $\tau(j) \neq \tau(j') \forall j \neq j'$

$$WATE = \frac{\sum_{j=1}^J \omega(j) \tau(j)}{\sum_{j=1}^J \omega(j)}$$

- $\omega(j) = \frac{N_j}{N} \frac{N_t(j)}{N(j)} \frac{N(j) - N_t(j)}{N(j)} = q(j) e(j) (1 - e(j))$

Asymptotic Inference

- The asymptotic variance of the WATE is

$$\mathbb{V}^{\text{strat}} = \frac{\sum_{i=1}^N \epsilon_i^2 \cdot \left(W_i - \sum_{j=1}^J q(j) B_i(j) \right)^2}{\left(\sum_{j=1}^J \omega(j) \right)^2}$$

⇒ Variance weights more those $\hat{\tau}(j)$ that are more precisely estimated

$$\mathbb{V}^j = \frac{\sigma^2 / N}{q(j)e(j)(1 - e(j))}$$

Fully-interacted Model

$$Y_i^{\text{obs}} = \tau W_i \frac{B_i(j)}{N(j)/N} + \sum_{j=1}^J \beta(j) B_i(j) + \sum_{j=1}^{J-1} \gamma(j) W_i \left(B_i(j) - B_i(J) \frac{N(j)}{N(J)} \right) + \epsilon_i$$

⇒ In this case, $\hat{\tau}^{\text{ols}}$ is a consistent estimator of ATE

- With asymptotic variance

$$\mathbb{V}^{\text{strat,inter}} = \sum_{i=1}^N q(j)^2 \cdot \left(\frac{\sigma_c^2(j)}{q(j)(1-e(j))} + \frac{\sigma_t^2(j)}{q(j)e(j)} \right)$$

⇒ In general, $\mathbb{V}^{\text{strat,inter}} > \mathbb{V}^{\text{strat}}$

Regression Analysis of the Tennessee Project Star

- The point estimate of τ in the standard model is

- $\hat{\tau}^{\text{ols}} = 0.238$ ($\widehat{s.e.} = 0.103$)

- The point estimate of τ in the fully-interacted model is

- $\hat{\tau}^{\text{ols,inter}} = 0.241$ ($\widehat{s.e.} = 0.095$)

⇒ Limited heterogeneity in the treatment effects across strata

Clustered Randomized Experiments

The Structure of Clustered Experiments

- Let G_{ig} be a binary indicator that unit i belongs to cluster $g = 1, \dots, G$
- $N_g = \sum_{i=1}^N G_{ig}$, so that N_g/N is the share of cluster g in the sample
- $\overline{W}_g \in \{0, 1\}$ is the **treatment assignment for all units** in cluster g
- The assignment mechanism is

$$P(\mathbf{W}, \mathbf{Y}(0), \mathbf{Y}(1), \mathbf{X}) = \binom{G}{G_t}^{-1}$$

$$\Rightarrow \mathbb{W}^+ = \{\mathbf{W} \in \mathbb{W} \mid \sum_{g=1}^G \overline{W}_g = G_t\}$$

Example: The *Progresa* Program (Again!)

- Clustered RCT during the roll-out of the program in rural areas
 - 506 villages among those eligible to receive the program
 - 320 early treatment and 186 late treatment (control)
- Individual/HH level data for both eligible and non-eligible HHs in each village
 - Approx. 30,000 program eligible children
 - About 50-100 HHs per village

Estimands

- The choice of **estimand depends on the choice of the unit** of analysis

⇒ **Unit-level**: a natural estimand is the usual ATE

$$\tau = \frac{1}{N} \sum_{i=1}^N (Y_i(1) - Y_i(0))$$

⇒ **Cluster-level**: weighted average of the unit-level treatment effect

$$\tau^{\text{clust}} = \frac{1}{G} \sum_{g=1}^G \tau_g, \quad \text{where } \tau_g = \frac{1}{N_g} \sum_{i: G_{ig}=1}^N (Y_i(1) - Y_i(0))$$

Unit-level Vs. Cluster-level

- Cluster-level analysis is more directly linked to the randomization framework
 - ⇒ Differently-sized clusters, such as states or towns
 - ⇒ Many units will be in the same treatment group so unit-level inference is tricky
- Unit-level incorporates individual-level covariates, which improve efficiency
 - ⇒ More homogenous clusters, such as schools or classrooms

Randomization Inference

- The usual statistic for unit-level analysis

$$T^{\text{dif}} = | \bar{Y}_t^{\text{obs}} - \bar{Y}_c^{\text{obs}} |$$

- The equivalent statistic for cluster-level analysis

$$T^{\text{clust}} = \left| \frac{1}{G_t} \sum_{g: \bar{W}_g=1} \bar{Y}_g^{\text{obs}} - \frac{1}{G_c} \sum_{g: \bar{W}_g=0} \bar{Y}_g^{\text{obs}} \right|$$

Randomization Inference of *Progresa*

- Children-level analysis on school enrollment (pre-program year 1997)

$$T^{\text{dif}} = 0.0075, \quad p\text{-value} = 0.400$$

- Children-level analysis on school enrollment (program year 1998)

$$T^{\text{dif}} = 0.0388, \quad p\text{-value} < 0.001$$

- Village-level analysis on school enrollment (program year 1998)

$$T^{\text{clust}} = 0.0234, \quad p\text{-value} = 0.0120$$

Regression Methods

- In unit-level analysis, we estimate the following regression

$$Y_i^{\text{obs}} = \alpha + \tau W_i + \bar{X}'\gamma + \epsilon_i$$

$\Rightarrow \hat{\tau}^{\text{ols}}$ consistently estimates ATE (with centered covariates)

- Use Liang-Zeger clustered Var-Cov:

$$\mathbb{V}_{\text{clust}} = \frac{\sum_{g=1}^G \left(\sum_{i:G_{ig}=1} \epsilon_i^2 \cdot (W_i - \bar{W})^2 \right)}{\left(\sum_{i=1}^N (W_i - \bar{W})^2 \right)^2}$$

Regression Analysis: Cluster-Level

- In cluster-level analysis, consider the following regression

$$\bar{Y}_g^{\text{obs}} = \alpha + \tau \bar{W}_g + \eta_g$$

$\Rightarrow \hat{\tau}^{\text{ols}}$ consistently estimates ATE

- The sampling variance of τ^{ols} is the usual one

$$\mathbb{V} = \frac{\sum_{g=1}^G \eta_g^2}{\sum_{g=1}^G (\bar{W}_g - \bar{W})^2} = \sigma^2 \left\{ \frac{1}{G_t} + \frac{1}{G_c} \right\}$$

Regression Analysis of *Progreso*

- Children-level analysis on school enrollment (pre-program year 1997)

$$\hat{\tau}^{\text{ols}} = 0.0075 (\widehat{s.e.} = 0.0091)$$

- Children-level analysis on school enrollment (program year 1998)

$$\hat{\tau}^{\text{ols}} = 0.0388 (\widehat{s.e.} = 0.0104)$$

- Village-level analysis on school enrollment (program year 1998)

$$\hat{\tau}^{\text{ols}} = 0.0234 (\widehat{s.e.} = 0.0092)$$

Dynamic Latent Factor Models

The Production Function of Human Capital

$$\mathbf{H}_{t+1} = g_t(\mathbf{H}_t, \mathbf{X}_t, \mathbf{I}_{t+1}, u_{t+1})$$

- $\mathbf{H}_t = \{\theta_{c,t}, \theta_{s,t}, \theta_{h,t}\}$ is human capital (including cognition, socio-emotional, and health)
- $\mathbf{X}_t = \{\theta_{m,t}, \theta_{f,t}, \theta_{r,t}\}$ are parental skills and other background variables (including mother, father and other caregivers)
- $\mathbf{I}_{t+1} = \{\theta_{M,t}, \theta_{T,t}\}$ are parental investments (including materials and time) between t and $t + 1$
- Stocks of current period skills produce next period skills

Self-Productivity and Dynamic Complementarities

- Self-productivity arises when

$$\frac{\partial g_t}{\partial \mathbf{H}_t} > 0$$

- Period t stocks of abilities and skills promote the acquisition of skills by making investment more productive.

$$\frac{\partial^2 g_t}{\partial I_t \partial \mathbf{H}_t} > 0$$

- Such complementarity explains why returns to educational investments are higher at later stages of the child life cycle for more able children

Measurement System

- For each $k \in \{c, n, m, f, \dots\}$ and period t we have measures such that:

$$m_{k,t,j} = \alpha_{k,t,j} \theta_{k,t} + \epsilon_{k,t,j}$$

- $m_{k,t,j}$ is the j^{th} measurement of skill $\theta_{k,t}$
- $\alpha_{k,t,j}$ is a factor loading
- $\epsilon_{k,t,j}$ is the measurement error contained in $m_{k,t,j}$
- With two independent measures per factor, the distribution of the unobserved factor and the measurement error can be identified non-parametrically up to a change of location
 - To set the factor scale: $\alpha_{k,t,1} = 1, \forall t = 1, \dots, T$
 - To set the factor location: $E(\theta_{k,t}) = 0, \forall t = 1, \dots, T$

Identification: Factor Loadings

- Assume that the $\epsilon_{k,t,j}$ are uncorrelated across t , then

$$\text{Cov}(m_{k,t,1}, m_{k,t+1,1}) = \text{Cov}(\theta_{k,t}, \theta_{k,t+1})$$

$$\text{Cov}(m_{k,t,2}, m_{k,t+1,1}) = \alpha_{k,t,2} \text{Cov}(\theta_{k,t}, \theta_{k,t+1})$$

- Hence, we can identify the factor loading

$$\frac{\text{Cov}(m_{k,t,1}, m_{k,t+1,1})}{\text{Cov}(m_{k,t,2}, m_{k,t+1,1})} = \alpha_{k,t,2}$$

- With more than two measures per factor, this approach can be generalized to allow for serial correlation in the $\epsilon_{k,t,j}$
 - Need $2M + 1$ measures to identify factor loadings for M factors

Identification: Distribution of Latent Variables

- Once $\alpha_{k,t,j}$ are identified, we can re-write the measurement system as

$$\frac{m_{k,t,j}}{\alpha_{k,t,j}} = \theta_{k,t} + \frac{\epsilon_{k,t,j}}{\alpha_{k,t,j}}$$

- The signal is common to multiple measurements, but the noise is not
- To extract the noise from the signal, assume that the $\epsilon_{k,t,j}$:
 - ① Are uncorrelated across measures conditional on θ
 - ② Are independent from factors
- Then we can identify the joint distribution of $\{\theta_{k,t}\}_{t=1}^T$
- Cunha et al (2010) also discuss identification under the more general case with unknown mapping between measures and factors and non-separable measurement errors

The Investment Problem

- Parents choose investments by solving

$$\begin{aligned} \max_{\{C_t, I_{t+1}\}} & U(C_t, \mathbf{H}_{t+1}) \\ \text{s.t.} & C_t + \mathbf{P}_t^I I_{t+1} = Y_t \\ & \mathbf{H}_{t+1} = g_t(\mathbf{H}_t, \mathbf{X}_t, \mathbf{I}_{t+1}, u_{t+1}) \end{aligned}$$

- Investment could be in commodities or in time
- Could easily add parental labor supply decisions
- Could easily add other inter-temporal dimensions beyond HK

Endogeneity of Investment

- Parents choose investment in reaction to shocks, and so $E(u_{t+1}|I_{t+1}) \neq 0$
- Posit a reduced-form of the investment function

$$I_{t+1} = f_t(Y_t, P_t, X_t, e_{t+1})$$

- Notice that Y_t and P_t are excluded from the production function
- One can estimate the investment equation jointly with the production function
- Or use a control function approach to estimate the role of investments in the production function
 - Plug estimated residuals $\hat{e}_{t+1}$ as additional covariates in the production function, so that $E(u_{t+1}|I_{t+1}, \hat{e}_{t+1}) = 0$

Estimation

- Estimation can be performed in two-steps
 - 1 Estimate the joint distribution of all latent factors
 - 2 Draws from such a distribution to estimate the investment equation as well as the production function with control function terms
- Inference can be done by using appropriate bootstrapping
 - This takes into account both estimation error at each stage and simulation error

The production of human capital in the early years

Estimating the Production Function for Human Capital: Results from a Randomized Controlled Trial in Colombia

- Human capital formation is a complex process
 - Human capital is **multi-dimensional** (cognitive, non-cognitive, health,...)
 - Skill formation is a **dynamic** process
 - The components of human capital **interact both within and across periods**
 - Both **genetic and environmental factors** are important inputs
- ⇒ The **early years** is a particularly salient period for policy
- Human capital is malleable (and vulnerable)
 - Dynamic complementarities (skills beget skills)
 - There is no equity-efficiency trade-off for early investment

The Intervention and Its Evaluation

- Adapt a prominent [ECD program in Jamaica](#) to the Colombian context
 - Children 12-24 months living in families targeted by a cash transfer program
 - Weekly home visits of one hour from trained women from the community
 - Micronutrients supplementation
 - Cheap and scalable intervention

⇒ [Clustered RCT design, stratified by region](#)

- 1,429 children in 96 municipalities in central Colombia
- Municipalities were [randomized into 4 groups](#) (cross-cutting or factorial design)

⇒ Control

⇒ Stimulation (home visits)

⇒ Supplementation (micronutrients)

⇒ Stimulation+Supplementation

Rich Data Collected on Children and Parents

- A battery of **psychometrics on children development** (baseline and follow-up)
 - ⇒ Motor and cognitive skills: Bayley test
 - ⇒ Socio-emotional skills: Bates temperament test
 - ⇒ Language development: MacArthur-Bates test
 - ⇒ Height, weight, hemoglobin, and morbidity
 - ⇒ Food intakes
 - ⇒ Child care arrangements and time use
- Mothers and families
- Home visitor
- Qualitative instruments

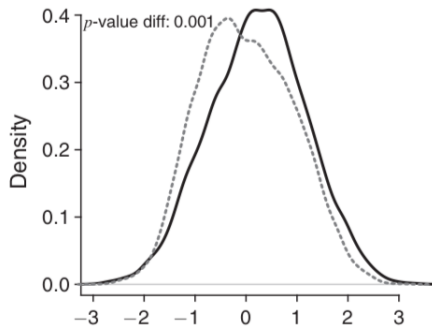
Experimental Results

⇒ Stimulation improved cognitive scores by 0.25σ + receptive language by 0.18σ

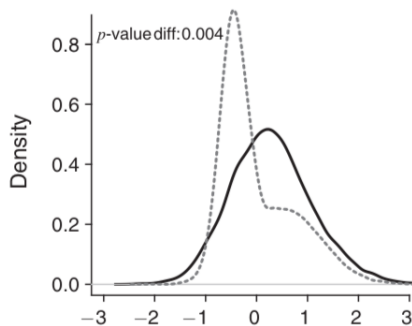
- Also positive (noisy) effects on socio-emotional skills
- Micronutrient supplementation had no significant effect on any outcome
- No effect of the intervention on height, weight, or hemoglobin levels

A First Hint at the Mechanisms: Parental Investment

Panel E. Material investments, follow-up



Panel F. Time investments, follow-up



— Treated ---- Control

What is the Rationale for the Model?

- Intervention changes human capital through **three different channels**
 - ⇒ Total factor productivity
 - ⇒ The productivity of the individual inputs
 - ⇒ The distribution of inputs (and in particular parental investment)
- Jointly estimate the **distribution of inputs and the production function**
- Unpack the role of these different hypotheses

Measurement Equations

$$m_{j,d}^{\theta} = \mu_{j,d}^k + \alpha_j^k \ln \theta_d^k + \epsilon_j^k$$

- Treated and control children are allowed to have different factor distributions
- Latent factors are distributed as a mixture of two normal distributions
- Measurement errors terms are jointly normal
- Likelihood function for each group writes as

$$L_i(m) = \int_{\theta} f(m_i|\theta) [\tau \phi_A(\theta) + (1 - \tau) \phi_b(\theta)] d\theta$$

$\Rightarrow f(m_i|\theta)$ is the density of the measurement system

Signal-to-noise Ratios:
$$s_{j,d}^{ln\theta^k} = \frac{(\alpha_j^k)^2 var(ln\theta^k)}{(\alpha_j^k)^2 var(ln\theta^k) + var(\epsilon_j^k)}$$

Latent factor	Measurement	Survey	Control	Treatment
Child's cognitive skills at FU (θ_{t+1}^C)	Bayley: Cognitive	FU	0.78	0.79
	Bayley: Receptive language	FU	0.75	0.76
	Bayley: Expressive language	FU	0.78	0.79
	Bayley: Fine motor	FU	0.59	0.61
	MacArthur: Words the child can say	FU	0.64	0.65
	MacArthur: Complex phrases the child can say	FU	0.52	0.54
Child's cognitive skills at BA (θ_t^C)	Bayley: Cognitive	BA	0.70	0.70
	Bayley: Receptive language	BA	0.73	0.72
	Bayley: Expressive language	BA	0.75	0.74
	Bayley: Fine motor	BA	0.60	0.59
	MacArthur: Words the child can say	BA	0.45	0.44
Child's socio-emotional skills at FU (θ_{t+1}^S)	ICQ: Difficult (-)	FU	0.74	0.71
	ICQ: Unsociable (-)	FU	0.33	0.30
	ICQ: Unstoppable (-)	FU	0.59	0.55
	ECBQ: Inhibitory control	FU	0.73	0.69
	ECBQ: Attentional focusing	FU	0.27	0.24
Child's socio-emotional skills at BA (θ_t^S)	ICQ: Difficult (-)	BA	0.68	0.71
	ICQ: Unsociable (-)	BA	0.28	0.31
	ICQ: Unadaptable (-)	BA	0.35	0.38
	ICQ: Unstoppable (-)	BA	0.22	0.25

Treatment Impacts on Measures and Factors

	Treatment effect		
	Point estimate	SE	Sample size
<i>Panel A. Child's cognitive skills at follow-up</i>			
Bayley: Cognitive	0.250	(0.063)	1,264
Bayley: Receptive language	0.175	(0.063)	1,264
Bayley: Expressive language	0.032	(0.062)	1,263
Bayley: Fine motor	0.072	(0.060)	1,262
MacArthur: Words the child can say	0.092	(0.064)	1,322
MacArthur: Complex phrases the child can say	0.058	(0.055)	1,322
Cognitive factor	0.115	(0.051)	
<i>Panel B. Child's socio-emotional skills at follow-up</i>			
ICQ: Difficult (-)	0.074	(0.045)	1,326
ICQ: Unsociable (-)	0.041	(0.054)	1,326
ICQ: Unstoppable (-)	0.032	(0.054)	1,326
ECBQ: Inhibitory control	-0.003	(0.058)	1,323
ECBQ: Attentional focusing	0.069	(0.049)	1,323
Socio-emotional factor	0.087	(0.044)	
<i>Panel C. Material investments at follow-up</i>			
FCI: Number of types of play materials	0.215	(0.064)	1,326
FCI: Number of coloring and drawing books	-0.133	(0.056)	1,326
FCI: Number of toys to learn movement	-0.048	(0.065)	1,326
FCI: Number of toys to learn shapes	0.416	(0.088)	1,326
FCI: Number of shop-bought toys	0.024	(0.061)	1,326
Material investment factor	0.227	(0.069)	
<i>Panel D. Time investments at follow-up</i>			
FCI: Number of types of play activities in last 3 days	0.277	(0.050)	1,326
FCI: Number of times told a story to child in last 3 days	0.138	(0.060)	1,326
FCI: Number of times read to child in last 3 days	0.362	(0.062)	1,326
FCI: Number of times played with toys in last 3 days	0.175	(0.060)	1,326
FCI: Number of times named things to child in last 3 days	0.137	(0.048)	1,326
Time investment factor	0.302	(0.068)	

The Production Function for Human Capital

- Cobb-Douglas specification for the technology of skill formation

$$\ln(\theta_{i,d,t+1}^k) = A_d^k + \gamma_{1,d}^k \ln(\theta_{i,t}^C) + \gamma_{2,d}^k \ln(\theta_{i,t}^S) + \gamma_{3,d}^k \ln(P_{i,t}^C) + \gamma_{4,d}^k \ln(P_{i,t}^S) \\ + \gamma_{5,d}^k \ln(I_{i,t+1}^M) + \gamma_{6,d}^k \ln(I_{i,t+1}^T) + \gamma_{7,d}^k \ln(n_{i,t}) + \eta_{i,t+1}^k$$

- A_d^k captures the direct effect of the home-visitor during her weekly visit

⇒ All parameters are skill-specific and depend on treatment status

Parental Investments

- Same log-linear specification for investment equation ($\tau = \{M, T\}$)

$$\begin{aligned}\ln(I_{i,t+1}^\tau) = & \lambda_{0,d}^\tau + \lambda_{1,d}^\tau \ln(\theta_{i,t}^C) + \lambda_{2,d}^\tau \ln(\theta_{i,t}^S) + \lambda_{3,d}^\tau \ln(P_i^C) \\ & + \lambda_{4,d}^\tau \ln(P_i^S) + \lambda_{5,d}^\tau \ln(n_{i,t}) + \lambda_{6,d}^\tau Z_{i,t} + u_{i,t+1}\end{aligned}$$

- Z_t are instrument
- All parameters are allowed to vary with the treatment
- **Control function** to account for correlation between shocks and investments

⇒ Include residuals $\hat{u}_{i,t+1}$ as additional regressor in the production function

Evidence Supporting the Exclusion Restrictions

TABLE 3—BALANCE TEST FOR THE INSTRUMENTAL VARIABLES

	log toy price	log food price	Conflict
Mother's cognitive skill	0.021 (0.009)	0.007 (0.007)	−0.004 (0.003)
Mother's socio-emotional skill	−0.006 (0.008)	−0.014 (0.006)	0.005 (0.005)
Mother is married	0.013 (0.016)	0.030 (0.012)	0.017 (0.008)
Wealth index	−0.003 (0.007)	0.002 (0.006)	0.002 (0.004)
Terrorism	0.015 (0.014)	−0.012 (0.013)	0.005 (0.009)
Constant	8.025 (0.028)	8.062 (0.017)	0.049 (0.007)
Observations	1,010	1,023	1,023
R^2	0.011	0.021	0.017
F -statistic	1.656	2.147	1.438
F -test p -value	0.154	0.0671	0.219

Notes: All right-hand-side variables measured at baseline. Asymptotic standard errors in parentheses allowing for clustering at the municipality level. *Terrorism* is the number of terrorist attacks between conception of child and baseline. *Conflict* is the number of conflicts against civil population divided by the municipality population (in thousands) when the mother herself was a child.

Estimation in Three Steps

- ⇒ Estimate the **cov. of the factors** based on cov. of the observed measures
- ⇒ Use these estimates to **predict factor scores**: $\hat{\theta}_{i,t}^k, \hat{\theta}_{i,t}^k$
- ⇒ Use predicted scores as data to **estimate the parental inv. and prod. function**
 - Compute conf. intervals for test statistics using the **block bootstrap**

Investment Functions

TABLE 5—ESTIMATES OF THE MATERIAL AND TIME INVESTMENT EQUATIONS

	Instruments: prices and conflict		Instruments: prices only
	Material investment	Time investment	Material investment
Intercept	−0.015 [−0.114, 0.078]	0.001 [−0.089, 0.089]	−0.013 [−0.11, 0.078]
Treatment	0.209 [0.038, 0.365]	0.318 [0.155, 0.48]	0.204 [0.037, 0.362]
log child's cognitive skill (t)	0.130 [0.016, 0.246]	0.068 [−0.044, 0.18]	0.132 [0.017, 0.25]
log child's socio-emotional skill (t)	−0.028 [−0.133, 0.087]	0.027 [−0.083, 0.145]	−0.030 [−0.131, 0.088]
log mother's cognitive skill	0.748 [0.582, 0.939]	0.349 [0.162, 0.498]	0.750 [0.583, 0.943]
log mother's socio-emotional skill	0.069 [−0.008, 0.139]	0.022 [−0.06, 0.108]	0.068 [−0.008, 0.139]
log number of children	−0.129 [−0.18, −0.077]	−0.128 [−0.186, −0.072]	−0.128 [−0.18, −0.078]
log toy price	−0.096 [−0.168, −0.027]	−0.020 [−0.085, 0.037]	−0.094 [−0.163, −0.026]
log food price	0.091 [0.006, 0.178]	0.042 [−0.026, 0.121]	0.091 [0.006, 0.178]
Maternal childhood exposure to conflict	−0.009	−0.089	

Production Function of Cognitive Skills

TABLE 6—ESTIMATES OF THE PRODUCTION FUNCTION FOR COGNITIVE SKILLS

Instruments:	OLS	IV			
	(1)	Prices, conflict (2)	Prices, conflict, treatment (3)	Prices (4)	Prices and treatment (5)
Intercept	−0.018 [−0.094, 0.053]	−0.019 [−0.111, 0.079]	0.007 [−0.089, 0.104]	0.003 [−0.091, 0.072]	−0.009 [−0.083, 0.058]
Treatment	0.083 [−0.025, 0.192]	0.049 [−0.12, 0.391]		−0.028 [−0.186, 0.156]	
log child's cognitive skill (<i>t</i>)	0.675 [0.589, 0.77]	0.648 [0.544, 0.795]	0.638 [0.522, 0.747]	0.626 [0.525, 0.746]	0.631 [0.533, 0.739]
log child's socio-emotional skill (<i>t</i>)	0.001 [−0.091, 0.087]	0.012 [−0.098, 0.143]	0.019 [−0.094, 0.14]	0.015 [−0.084, 0.126]	0.02 [−0.079, 0.127]
log mother's cognitive skill	0.213 [0.089, 0.35]	−0.075 [−0.456, 0.5]	−0.173 [−0.538, 0.201]	−0.102 [−0.495, 0.291]	−0.094 [−0.45, 0.21]
log mother's socio-emotional skill	0.103 [0.031, 0.173]	0.084 [−0.03, 0.163]	0.063 [−0.035, 0.151]	0.06 [−0.019, 0.152]	0.074 [−0.026, 0.155]
log number of children	0.042 [−0.01, 0.092]	0.085 [−0.07, 0.154]	0.084 [0.011, 0.163]	0.089 [0.002, 0.176]	0.086 [0.023, 0.164]
log material investment	0.088 [0.016, 0.157]	0.594 [0.025, 1.179]	0.784 [0.204, 1.383]	0.542 [0.041, 0.996]	0.516 [0.195, 0.946]
log time investment	0.038 [−0.051, 0.129]	−0.171 [−1.198, 0.312]	−0.311 [−0.985, 0.217]		

Goodness-of-fit: Gap in output between treated and control

Measured in the data

0.115

0.115

0.115

0.115

0.115

Production Function of Socio-emotional Skills

TABLE 7—ESTIMATES OF THE PRODUCTION FUNCTION FOR SOCIO-EMOTIONAL SKILLS

Instruments:	OLS	IV			
	(1)	Prices, conflict (2)	Prices, conflict (3)	Prices, conflict, treatment (4)	Prices, conflict, treatment (5)
Intercept	−0.009 [−0.08, 0.063]	−0.02 [−0.087, 0.064]	−0.006 [−0.084, 0.066]	−0.02 [−0.096, 0.058]	−0.022 [−0.089, 0.051]
Treatment	−0.011 [−0.124, 0.093]	−0.071 [−0.318, 0.166]	−0.116 [−0.31, 0.115]		
log child's cognitive skill (<i>t</i>)	0.106 [0.018, 0.192]	0.074 [−0.019, 0.21]	0.09 [−0.017, 0.194]	0.094 [0.003, 0.22]	0.099 [0.002, 0.193]
log child's socio-emotional skill (<i>t</i>)	0.522 [0.403, 0.659]	0.513 [0.374, 0.672]	0.499 [0.389, 0.663]	0.516 [0.387, 0.669]	0.512 [0.403, 0.656]
log mother's cognitive skill	−0.077 [−0.217, 0.049]	−0.146 [−0.443, 0.297]	−0.15 [−0.349, 0.084]	−0.018 [−0.315, 0.328]	−0.083 [−0.231, 0.078]
log mother's socio-emotional skill	0.037 [−0.058, 0.119]	0.035 [−0.062, 0.141]	0.048 [−0.053, 0.135]	0.049 [−0.054, 0.134]	0.043 [−0.042, 0.126]
log of number of children	0.099 [0.047, 0.153]	0.127 [0.017, 0.236]	0.133 [0.034, 0.223]	0.101 [0.028, 0.168]	0.101 [0.036, 0.163]
log material investment	0.154 [0.06, 0.256]	0.015 [−0.621, 0.428]		−0.144 [−0.688, 0.34]	
log time investment	0.109 [−0.006, 0.213]	0.487 [−0.177, 1.258]	0.549 [−0.022, 1.147]	0.448 [−0.125, 1.057]	0.324 [0.025, 0.691]

Goodness-of-fit: Gap in output between treated and control

Measured in the data

0.087

0.087

0.087

0.087

0.087

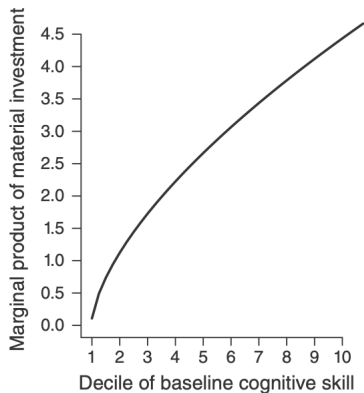
Interpreting the Impact of the Intervention

- Cognitive (Socio-emotional) development is mainly determined by
 - Initial cognition (measured at ages 1-2)
 - Stimulation provided by older siblings
 - **Material (time) investments** provided by parents

⇒ The main channel of the intervention is a shift in parental investments

Complementarities between Inputs

Panel A. Production function for cognitive skills



Panel B. Production function for socio-emotional skills

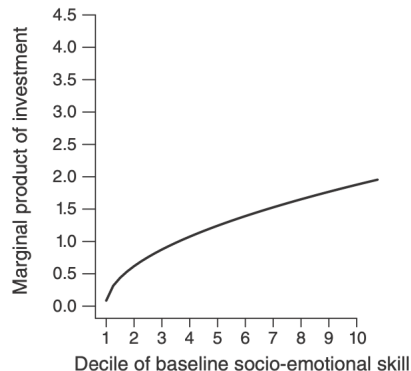


FIGURE 2. COMPLEMENTARITY BETWEEN INVESTMENTS AND BASELINE SKILLS

Implications for longer-term outcomes

- Parental crowding-in suggests treatment will diverge in the medium/long run
- But also possible fade-out due to mean reversion
- Follow-up data (2 years after) suggests **partial fade out**

⇒ How can we permanently **improve parental practices** ?

RCT $\Leftrightarrow$ Economic Model

- Model gives a lens to **interpret how the intervention** affected child development

⇒ Sometimes the experimental variation is what you want to explain

- Need additional sources of variation beyond the experiment